Program : **Diploma in Printing Technology**	
Course Code : **4103**	Course Title: **Digital Imaging Techniques**
Semester : **4**	Credits: **4**
Course Category: **Program Core**	
Periods per week: **4 (L:3 T: 1 P: 0)**	Periods per semester: **60**

Course Objectives:

- To provide a cutting edge knowledge in Digital imaging technique.
- To keep pace with this changes taking place in the international field of Printing & Graphic Arts Technology.
- To expose the students in the fundamentals of Digital Prepress followed by the wide application of computers in digital printing, pre-press and reproduction technique.

Course Prerequisites:

Topic	Course code	Course Title	Semester
Bit, byte, bitmap images.		Introduction to IT system	1
WYSIWYG, Color theories		Fundamentals of Color Reproduction	2
Types of images, types of Originals		Print Design	3

Course Outcomes

On completion of the course ,the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Explain different file formats and their use proof and proofing methods, PDF its production and management.	15	Understanding
CO2	Interpret data transfer by using computer.	14	Understanding
CO3	Summarize the modern digital printing technologies.	15	Understanding
CO4	Outline the working of inkjet and laser printing technology.	14	Understanding
	Series Test	2	

CO - PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Explain different file formats and their use proof and proofing methods, PDF its production and management.**		
M1.01	Identify different file formats.	3	Understanding
M1.02	Identify the importance of PDF.	4	Understanding
M1.03	Explain digital imposition.	4	Understanding
M1.04	Explain RIP and its stages.	4	Understanding
Contents: Different File formats- their features, and uses in printing - JPEG/JPG, TIFF, PNG, GIF, BMP, PDF, JDF, EPS, PSD, CRD, DOC/DOCX, PPT/PPTX, FLASH-SWF, ZIP, HTML, QXD, PMD, INDD, SLA/SCD, TXT, ODT, ICC/ICM, XML. Creating/saving in different file formats with different application programs (eg, PSD - Photoshop, CDR - Corel draw, DOCX - MS Word etc). Differentiate Vector file formats and raster file formats. PDF: PDF - history, importance, advantages.PDF writer and Adobe Distiller. PDF in print industry. PDF and post script differences and similarities. JDF - definition, use, importance. Setting of a digital document - Art box, trim box, bleed box, media box. Trimming marks, cutting marks, quality control patches, registration marks etc. RIP (raster image processer): definition, function, hardware and software. Stages of RIP. Data transfer: Data transfer definition. Print data exchange. Data exchange standards (PDF, PDF-X series). Introduction to page description language. Names of page description language. Postscript language. Proofing methods: digital proofing. Digital halftone proof. Color and black and white materials.			
CO2	**Interpret data transfer by using computer.**		
M2.01	Define modern print workflow	3	Understanding
M2.02	Understand print workflow software.	4	Understanding
M2.03	Explain Job tickets	3	Understanding
M2.04	Describe about CTP technologies	4	Understanding
	Series Test I	1	

Contents: Digital print production workflow (automated workflow). Production management software. Production workflow systems - Prinect (signastation, Meta dimension etc.) Software for (names and manufacturers) - Preflight. Job ticket - definition, use, role in production. Digital job tickets. Processing information contained in the job tickets. Digital job ticket based work flow. CTP (computer to plate) technology. Types of CTP plates. Types of CTP systems - UV, Thermal. Working of CTP machine. Names of major manufacturers of CTP machines.			
CO3	**Summarize the modern digital printing technologies.**		
M3.01	State the concept of digital printing	3	Understanding
M3.02	Explain different digital printing technologies	4	Understanding
M3.03	Define impact and non impact printing	4	Understanding
M3.04	Identify the characteristics of Digital printing inks.	4	Understanding
Contents: Definition of digital printing. Basic digital printing work flow. Digital printing application. Advantages of digital printing - cost, consistency, storage, size, artistic control, freedom and flexibility. Digital image - Pixel and bit depth. Digital printing technologies - Electro photography, Ionography, Thermography, toner jet, Magnetography, Dot matrix printers, Inkjet Printers. Characteristics of Toner inks, Inkjet inks.			
CO4	**Outline the working of inkjet and laser printing technology.**		
M4.01	Define inkjet printing.	2	Understanding
M4.02	Explain the working principles of ink jet printing.	3	Understanding
M4.03	Define laser printing.	2	Understanding
M4.04	Define LED printer.	2	Understanding
M4.05	Identify different digital printing substrates.	2	Understanding
M4.06	Understand the working principle of direct imaging presses.	3	Understanding
	Series Test II	1	
Contents: Definition of ink jet printing. Classification of ink jet printing -Drop on demand ink jet, continuous inkjet. Brief explanation of working principles of - Binary deflection, multi deflection, thermal ink jet, piezo ink jet, electrostatic ink jet, solid ink. Cartridge - print head. Laser printing. Brief explanation of working principle of Laser printer. LED printer. Brief explanation of working principle of LED printer. Comparison of ink jet and laser printing, advantages and disadvantages. Digital printing substrates - paper, board, plastics, cloth, metal foils/sheets, glass, ceramic. Comparison between Digital printing and offset printing, advantages and disadvantages. Direct imaging technology. Direct imaging presses. Features of direct imaging presses. Plates in direct imaging press.			

Text / Reference:

T/R	Book Title/Author
R1	**Heidelberg Expert guide PDF workflow**. Heidelberg.
R2	Michael Bernard, John Peacock., **Handbook of printing and production,** Springer Science & Business.
R3	Helmut Kipphan, **Hand book of print media,** Springer Science & Business. Media
R4	Reid Anderson, **Exploring Digital Prepress: 1 edition**, Delmar Cengage Learning.
R5	Adams J.M, **Printing Technology 5th edition,** Delmar Publishers, NewYork
R6	Harald Jhonson, **Mastering Pigital Printing Second Edition.**

Online resources:

Sl.No	Website Link
1	https://en.wikipedia.org › wiki › Digital_imaging
2	https://www.springer.com › book